

CARBOXYLATED NBR: PATENT LANDSCAPE FOR SYNTHESIS AND PROCESSING

1. ABSTRACT

This report synthesizes two primary patents relevant to carboxylated nitrile butadiene rubber (NBR), focusing on both direct synthesis of carboxyl group-containing nitrile rubber and downstream use of carboxylated NBR in heat-activable adhesive systems. EP3604355A1 discloses a solvent-based polymerization route in which a monomer mixture containing an α,β -ethylenically unsaturated nitrile monomer and a carboxyl group-containing monomer is polymerized, then terminated with a specific combination of a nitrous acid salt and a water-insoluble hydroquinone before coagulation. The key technical objective is to obtain carboxylated nitrile rubber with reduced discoloration and stable storage behavior, while maintaining effective polymerization termination at low terminator dosage. US20100147462A1 is not a synthesis patent per se, but it is highly relevant because it uses acid- or acid anhydride-modified acrylonitrile-butadiene copolymers as the elastomeric component in heat-activable adhesive tapes for flexible printed circuit boards. It confirms the industrial importance of carboxylated NBR and provides formulation constraints that preserve flexibility, solvent processability, and high-temperature bond performance. Across the two patents, the dominant process themes are copolymerization with carboxyl-containing monomers, solvent/emulsion-style rubber production, controlled termination/coagulation, and formulation of carboxylated NBR with epoxy resins for thermally activated bonding. The evidence indicates that methacrylic acid or related carboxyl monomers are preferred for introducing functionality, while nitrous acid salt/hydroquinone termination chemistry is used to suppress discoloration and viscosity drift. In application-oriented formulations, the carboxylated NBR fraction is used as the reactive elastomeric hardener for epoxy resin, with acrylonitrile content in the 15% to 50% by weight range and the adhesive designed to crosslink above 150°C. Overall, the patents show that carboxylated NBR is valued both as a directly synthesized functional rubber and as a reactive polar elastomer for high-adhesion, heat-resistant adhesive systems.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3604355A1
- Patent Title: Method for producing carboxylated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses producing carboxyl group-containing nitrile rubber by polymerizing a monomer mixture of an α,β -ethylenically unsaturated nitrile monomer and a carboxyl group-containing monomer in a solvent. Polymerization is terminated at a predetermined conversion by adding a nitrous acid salt together with a water-insoluble hydroquinone, followed by coagulation of the resulting dispersion to isolate the solid rubber. - Nitrous acid salt amount: 0.15 parts by weight or less per 100 parts by weight of monomer mixture — Abstract/Claims - Water-insoluble hydroquinone amount: 0.1 to 1.0 part by weight per 100 parts by weight of monomer mixture — Claims/Examples discussion - Nitrous acid salt to water-insoluble hydroquinone ratio: 1:0.5 to 1:100 — Description - Polymerization conversion ratio after addition of polymerization terminator: 65% — Comparative Example 1 - Polymerization conversion ratio after addition of polymerization terminator: 80% — Comparative Example 1 - pH of latex of carboxyl group-containing nitrile rubber: 3.8 to 5.6 — Comparative Example 1 - Increase in polymerization conversion ratio after storage at 5°C: 0.4% to 16.7% — Comparative Example 1 - Increase in polymerization conversion ratio after storage at 50°C: 0.0% to 32.5% — Comparative Example 1 - Increase in Mooney viscosity after storage at 5°C: 0.1 to 17.0 — Comparative Example 1 - Increase in Mooney viscosity after storage at 50°C: 0.0 to 43 — Comparative Example 1 - Scorch stability (t5): 9.1 to 11 min — Comparative Example 1 - Monomer composition: acrylonitrile 25 parts, 1,3-butadiene 68 parts, methacrylic acid 7 parts — Comparative Example 1

Relevant Experimental Evidence - Comparative Example 1: Using nitrous acid salt plus water-insoluble hydroquinone at low dosage gave low post-storage conversion/viscosity increase and reduced discoloration. - Comparative Example 1: Alternative terminators produced higher storage instability and poorer discoloration performance. - Comparative Example 1: The data support the claimed terminator combination as necessary for effective polymerization termination and color control.

Technical Relevance This is the most directly relevant patent for carboxylated NBR synthesis because it explicitly teaches incorporation of a carboxyl-containing monomer into nitrile rubber and identifies a termination/coagulation strategy that improves product quality. It provides actionable synthesis guidance on monomer selection and terminator dosage for producing low-discoloration carboxylated NBR.

Patent 2

- Patent Number: US20100147462A1
- Patent Title: ... strip that can be activated by heat and is based on carboxylated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent does not disclose synthesis of carboxylated NBR; instead, it uses an acid- or acid anhydride-modified acrylonitrile-butadiene copolymer as the elastomeric component in a heat-activable adhesive tape. The modified nitrile rubber is dissolved with epoxy resin in butanone, coated onto a release paper, dried, and then used in lamination/bonding operations where the adhesive crosslinks under heat. - Nipol NX 775 loading: 80 parts by weight — Example 1 - Nipol NX 775 acrylonitrile content: 26% by weight — Example 1 - Nipol NX 775 acid modification: 7% by weight — Example 1 - Bakelite EPR 166 loading: 20 parts by weight — Example 1 - Bakelite EPR 166 epoxide equivalent: 184 — Example 1 - Coat thickness: 25 μm — Example 1 - Piccolyte A 125 loading: 20 parts by weight — Example 2 - Breon N41H80 loading: 75 parts by weight — Example 3 - Breon N41H80 acrylonitrile content: 41% by weight — Example 3 - Breon N41H80 Mooney viscosity ML 1+4 at 100°C: 72 to 88 — Example 3 - Bakelite EPR 166 loading: 25 parts by weight — Example 3 - Dyhard 100-S loading: 5 parts by weight — Example 4 - Bonding temperature: 100°C — Bonding procedure - Press temperature: 200°C — Bonding procedure - Press pressure: 1.3 MPa — Bonding procedure - Press time: 1 hour — Bonding procedure - T-peel test rate: 50 mm/min — Test methods - Test temperature: 20°C — Test methods - Relative humidity: 50% — Test methods - Static peel test temperature: 70°C — Test methods - Static peel load: 500 g — Test methods - Solder bath temperature: 288°C — Test methods - Solder bath exposure time: 10 seconds — Test methods - Acrylonitrile content of nitrile rubbers used: 15% to 50% by weight — Claims/Description - Adhesive crosslinking temperature: above 150°C — Claims

Relevant Experimental Evidence - Example 1: A carboxylated nitrile rubber/epoxy resin adhesive tape was prepared from Nipol NX 775 and showed strong bonding performance. - Example 2: Addition of polyterpene resin modified the adhesive formulation while retaining the same processing route. - Example 3: A higher-acrylonitrile nitrile rubber formulation was evaluated as a comparative system. - Example 4: A formulation with added dicyandiamide hardener was tested as a comparative reference. - Example 1-4: All samples passed the solder bath resistance test, while the acid-modified nitrile rubber systems showed superior flexibility/bond strength behavior versus unmodified reference systems.

Technical Relevance Although not a synthesis patent, this document is highly relevant because it demonstrates how carboxylated NBR functions as a reactive polar elastomer in industrial adhesive formulations. It helps define useful composition windows and processing conditions for downstream utilization of carboxylated NBR produced by synthesis routes like those in

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP3604355A1 is a direct synthesis patent for carboxylated nitrile rubber, while US20100147462A1 is an application/formulation patent using acid-modified NBR as a reactive adhesive component.
- EP3604355A1 explicitly incorporates a carboxyl group-containing monomer into the nitrile rubber backbone; US20100147462A1 relies on pre-existing acid- or acid anhydride-modified acrylonitrile-butadiene copolymer.
- EP3604355A1 emphasizes polymerization termination chemistry: nitrous acid salt plus water-insoluble hydroquinone, with nitrous acid salt at 0.15 parts by weight or less per 100 parts monomer mixture.
- US20100147462A1 emphasizes adhesive formulation ratios and thermal processing: elastomer/epoxy resin weight ratio greater than 1.5, no additional nonpolymer hardener, lamination at 100°C, pressing at 200°C and 1.3 MPa.
- The monomer composition in EP3604355A1 example data is acrylonitrile 25 parts, 1,3-butadiene 68 parts, and methacrylic acid 7 parts; US20100147462A1 cites commercial nitrile rubbers with acrylonitrile contents of 26% and 41% by weight and acid modification of 7% by weight in one example.
- EP3604355A1 reports latex pH values of 3.8 to 5.6 and storage-related conversion/viscosity changes as quality metrics; US20100147462A1 reports adhesive performance metrics such as T-peel, static peel at 70°C, and solder bath resistance.
- EP3604355A1 uses coagulation after termination to isolate the solid rubber; US20100147462A1 uses solution coating onto release paper followed by drying and later heat activation/crosslinking in the bondline.

4. CONCLUSION

The evidence supports two complementary routes for carboxylated NBR: direct synthesis by copolymerization with a carboxyl-containing monomer, and downstream use of acid-modified NBR as a reactive elastomer in epoxy-based heat-activable adhesives. For synthesis, the strongest evidence favors a nitrile/butadiene/methacrylic acid-type composition with controlled termination using a nitrous acid salt and water-insoluble hydroquinone, keeping nitrous acid salt at 0.15 parts by weight or less per 100 parts monomer mixture and hydroquinone at 0.1 to 1.0 part by weight. For application, carboxylated NBR is most useful when paired with epoxy resin at an elastomer/resin ratio greater than 1.5 and processed under heat above 150°C. Overall, the patents indicate that carboxylated NBR should be synthesized with careful termination and coagulation control to minimize discoloration and storage instability, then formulated as a polar, reactive

elastomer for high-adhesion, heat-resistant products.

5. REFERENCES

- EP3604355A1 | Method for producing carboxylated nitrile rubber | Not disclosed | EP | 2020 | <https://patents.google.com/patent/EP3604355A1/en>
- US20100147462A1 | ... strip that can be activated by heat and is based on carboxylated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US20100147462A1/en>